CIVE 481/881: Computational Problem Solving in Civil Engineering

Spring 2026

3-credit hour

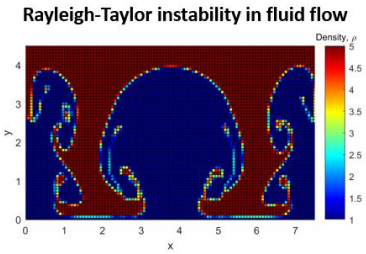
Available in Lincoln and Omaha

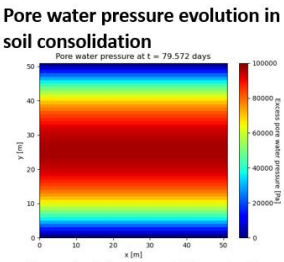
Monday and Wednesday 3:00 – 4:15 p.m.

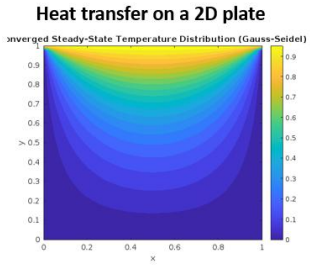
Pre-requisite: MATH 221 and CSCE 155A or 155E or 155H or 155N

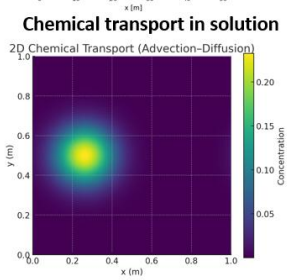
Instructor: Dr. Yumeng Zhao [yzhao52@unl.edu]

Course Overview: Do you wonder how chemicals migrate through water, how heat diffuses across solids, or how fluids become unstable and form striking patterns? Do you know how we simulate these physical processes inside a computer as if they were happening in the real world? This course reveals the mathematical foundations and computational techniques that make such predictions possible. Through a blend of theory and hands-on programming, you will learn how to model complex physical systems using both classical numerical methods and emerging AI-driven approaches. Tentative topics include:









- Numerical stability, consistency, convergence
- 1st order Ordinary Differential Equations (ODEs) (Euler-Forward / Backward / Centered Methods)
- Numerical linear algebra (Direct solvers; iterative solvers such as Jacobi / Gauss-Seidel / SOR Methods)
- Systems of ODEs
- Higher order ODEs (Runge-Kutta / Predictor-Corrector / Euler-Richardson Methods)
- 1st order, higher-order and systems of Partially Differential Equations (PDEs)
- Machine Learning for differential equations (Physical-Informed Neural Networks)
- Machine Learning for data-only tasks (Gaussian Processes / classification)

We will illustrate numerical methods using examples drawn from **solid mechanics, fluid dynamics, heat transfer, chemical transport, transportation networks, electromagnetics, biological systems, earth processes, and even human behavior and finance**. This course is designed for students across all fields of engineering and science seeking a theoretical and practical foundation in numerical modeling.

Student-Driven Study Plans: This course follows a student-centered study plan designed to emphasize exploration, creativity, and practical skill-building. Instead of a high-stakes final exam, students will complete **hands-on coding projects** that demonstrate their ability to simulate real physical systems using numerical methods. With only one midterm exam to assess core concepts, the majority of the course focuses on applying ideas through computation. Students will gain extensive experience using **MATLAB and Python** to translate mathematical models into simulations, empowering them to code the physics behind their own fields of interests.

Please contact Dr. Zhao at yzhao52@unl.edu with any questions about the course or enrollment.